

ZSCALER PRIVATE ACCESS LOG PARSER

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What is Zscaler Private Access?

Zscaler Private Access (ZPA) is a cloud-based Zero Trust Network Access (ZTNA) solution that securely connects users to internal applications without exposing them to the public internet. Unlike traditional VPNs, which provide broad network access, ZPA enforces application-specific access based on identity, device posture, and policies.

How do users Connect to the ZPA?

Step 1: User Requests Access

The user attempts to open a private application (e.g., hr.company.internal).

The Zscaler Client Connector (ZCC) on the user's device intercepts the request.

Step 2: Authentication & Policy Validation

The request is sent to the ZPA Cloud Broker via the closest ZEN Node (Zscaler Enforcement Node).

ZPA authenticates the user through an Identity Provider (Azure AD, Okta, etc.) and verifies security policies (Compliance, Patching, etc)

Step 3: App Connector Fetches the Application

If access is approved, ZPA selects the correct App Connector (a VM inside the corporate network or cloud).

The App Connector fetches the requested application's response and relays it back without exposing the internal network.

Step 4: Secure, Application-Level Access

The user never directly connects to the corporate network—only to the approved application via a temporary, encrypted TLS tunnel.

The session is continuously monitored, and access is dynamically revoked if risks are detected.

How are these logs collected in ZPA?

Zscaler Private Access (ZPA) continuously collects logs to provide visibility into user activity, application access, and security events. Logs are generated whenever a user attempts to access a private application through ZPA. These logs include Traffic Logs, Access Logs, and Audit Logs, capturing details such as user identity, source and destination IPs, ports, timestamps, and policy decisions. The logs are collected from multiple sources, including Zscaler Client Connector (ZCC), ZPA Cloud Broker, and App Connectors.

Forwarding ZPA Logs to a Syslog Server on an Amazon Linux EC2 Instance

Zscaler Private Access (ZPA) supports real-time log streaming to a Syslog server, allowing organizations to centralize security and access logs for monitoring, analysis, and compliance. To configure this, an Amazon Linux EC2 instance is set up as the Syslog server. First, rsyslog is installed and configured on the EC2 instance to listen for incoming logs on UDP or TCP port 514. The `/etc/rsyslog.conf` file is modified to enable remote logging, and the service is restarted to apply changes. On the ZPA Admin Portal, log streaming is configured under Administration > Log Streaming, where a new Log Receiver is added. The Syslog server's IP address, port (514), protocol (UDP/TCP), and log format (CEF/JSON) are specified. Once configured, ZPA forwards logs to the EC2 instance, where they can be stored, processed, or forwarded to a SIEM (e.g., Splunk, ELK) for deeper analysis. This setup ensures secure, centralized log management, providing visibility into user access, application activity, and security events within ZPA.

Log Parsing with ZPA

This Zscaler Private Access (ZPA) log captures essential details about a user session, providing visibility into user activity, connection status, and network details. The log records timestamps, identifying when the event occurred (2025-01-31T00:11:05.151014+00:00), along with the event type (User Activity) and source (zpa-lss).

It includes session-specific details, such as the customer identifier (XYZ), unique Session ID and Connection ID, and an internal reason code for tracking connection status. The log also indicates that the session was open and active, with a duration of 17 seconds and no data transferred at that moment.

The user identity and access details are recorded, including the user's email (vmusti@example.com), the application port (3389), and the source and destination IPs (192.0.2.10 → 203.0.113.12). The log further captures the user's geolocation (40.000000, -74.000000, USA) and the associated ZEN node (AP-USA-10001) handling the connection.

To track application access, the log lists the policy applied (XYZ-Region-Access), the App Connector (XYZ-DataCenter-Node1) facilitating the connection, and the internal server (10.0.1.2) hosting the requested application. The application name (XYZ-DataCenter-Desktops) and its segment group (XYZ-DataCenter-SegmentGroup) are also logged.

The log includes timestamps for various stages of the connection, such as when it started, when data was exchanged, and when it ended. Additionally, performance metrics, including policy processing time (177 ms) and server setup time (239 ms), help measure session efficiency.

Finally, traffic volume statistics are recorded, showing bytes sent and received by the client and connector (500000 bytes received, 1200 bytes transmitted). The log also identifies the Identity Provider (XYZ Azure AD), along with metadata like city (New York), micro-tenant details, and other attributes relevant to ZPA's access control and monitoring. You can find the sample log below.

SAMPLE ZPA-LSS LOG:

```
2025-01-31T00:11:05.151014+00:00 Tue Jan 31 00:11:05 2025 User Activity zpa-  
lss:,XYZ,AB1/CD2EF/3G4H5IJKLMN,AB1/CD2EF/3G4H5IJKLMN,XYZA0BCDE12345FGHIJ,O  
PEN_OR_ACTIVE_CONNECTION,active,17,0,vmusti@example.com,3389,192.0.2.10,203.0.11  
3.12,40.000000,-74.000000,USA,AP-USA-10001,XYZ-Region-Access,XYZ-DataCenter-  
Node1,AP-XYZ1234,10.0.0.1,47568,10.0.1.2,XYZ-DataCenter-Desktops,XYZ-DataCenter-  
SegmentGroup,0,10.0.1.2,3389,177,239,2025-05-04T22:56:05.343Z,,2025-05-  
04T22:56:05.469Z,2025-05-04T22:56:05.489Z,2025-05-04T22:56:05.368Z,2025-05-  
04T22:56:05.538Z,2025-05-04T22:56:05.496Z,2025-05-05T00:11:02.623Z,2025-05-  
05T00:11:02.610Z,2025-05-04T22:56:05.494Z,2025-05-04T22:56:05.496Z,2025-05-  
04T22:56:05.494Z,2025-05-05T00:11:02.610Z,2025-05-  
05T00:11:02.623Z,500000,1200,7500000,1500,7500000,1500,500000,1200, XYZ Azure  
AD,0, New York,0,0
```

To properly analyze and process ZPA logs, each entry should be parsed based on its structured format. The log fields should be extracted in the following order:

LogTimestamp,User Activity ,Customer,SessionID,ConnectionID,InternalReason,ConnectionSta
tus,IPProtocol,DoubleEncryption,Username,ServicePort,ClientPublicIP,ClientPrivateIP,ClientLat
itude,ClientLongitude,ClientCountryCode,ClientZEN,Policy,Connector,ConnectorZEN,Connecto
rIP,ConnectorPort,Host,Application,AppGroup,Server,ServerIP,ServerPort,PolicyProcessingTim
e,ServerSetupTime,TimestampConnectionStart,TimestampConnectionEnd,TimestampCATx,Ti
mestampCARx,TimestampAppLearnStart,TimestampZENFirstRxClient,TimestampZENFirstTxC
lient,TimestampZENLastRxClient,TimestampZENLastTxClient,TimestampConnectorZENSetup
Complete,TimestampZENFirstRxConnector,TimestampZENFirstTxConnector,TimestampZENLa
stRxConnector,TimestampZENLastTxConnector,ZENTotalBytesRxClient,ZENBytesRxClientZE
NTotalBytesTxClient,ZENBytesTxClient,ZENTotalBytesRxConnector,ZENBytesRxConnecto
r,ZENTotalBytesTxConnector,ZENBytesTxConnector,Idp,c2c,ClientCity,MicroTenantID,AppMicr
oTenantID.

Constructing the decoder

This corresponding decoder resides on the Wazuh server in the path **/var/ossec/etc/decoders**. It is this path in the custom decoders which will be used by the wazuh 's configuration (**/var/ossec/etc/ossec.conf**) server to decode the incoming logs. Therefore, any custom decoder should be present in this path for the configuration to be used.

In the Wazuh, we need to write a custom decoder, which parses the logs from the zpa-lss. Therefore, a custom decoder is written in XML, where primarily, a prematch is established by giving a decoder name.

Secondly, regex patterns are used to match the data in the logs and fetch them. Thirdly, an `<order>` tag is created so that the parsed values are matched according to the corresponding field names.

```
<decoder name="zpa-1ss">
```

```
<prematch>User Activity zpa-lss:</prematch>
```

```
<regex
```

[illegible]

</decoder>

Below is how the log is expected to appear in the SIEM after parsing:

LogTimestamp=2025-01-31T00:11:05.151014+00:00

DateTime=Tue Jan 31 00:11:05 2025

UserActivity=User Activity

EventSource=zpa-lss:

Customer=XYZ

SessionID=AB1/CD2EF/3G4H5IJKLMN

ConnectionID=AB1/CD2EF/3G4H5IJKLMN

InternalReason=XYZA0BCDE12345FGHIJ

ConnectionStatus=OPEN OR ACTIVE CONNECTION

ConnectionState=active

SessionDuration=17

DataTransferred=0

ClientUsername=vmusti@example.com

ApplicationPort=3389

ClientIP=192.0.2.10

PublicIP=203.0.113.12

ClientLatitude=40.000000

ClientLongitude=-74.000000

ClientCountryCode=USA

ClientZEN=AP-USA-10001

Policy=XYZ-Region-Access

Connector=XYZ-DataCenter-Node1

ConnectorZEN=AP-XYZ1234

ConnectorIP=10.0.0.1

ConnectorPort=47568

Host=10.0.1.2

Application=XYZ-DataCenter-Desktops

AppGroup=XYZ-DataCenter-SegmentGroup

Server=0

ServerIP=10.0.1.2

ServerPort=3389
PolicyProcessingTime=177
ServerSetupTime=239
TimestampConnectionStart=2025-05-04T22:56:05.343Z
TimestampConnectionEnd=2025-05-04T22:56:05.469Z
TimestampCATx=2025-05-04T22:56:05.489Z
TimestampCARx=2025-05-04T22:56:05.368Z
TimestampAppLearnStart=2025-05-04T22:56:05.538Z
TimestampZENFirstRxClient=2025-05-04T22:56:05.496Z
TimestampZENFirstTxClient=2025-05-05T00:11:02.623Z
TimestampZENLastRxClient=2025-05-05T00:11:02.610Z
TimestampZENLastTxClient=2025-05-04T22:56:05.494Z
TimestampConnectorZENSetupComplete=2025-05-04T22:56:05.496Z
TimestampZENFirstRxConnector=2025-05-05T00:11:02.610Z
TimestampZENFirstTxConnector=2025-05-05T00:11:02.623Z
TimestampZENLastRxConnector=2025-05-04T22:56:05.494Z
TimestampZENLastTxConnector=2025-05-04T22:56:05.496Z
ZENTotalBytesRxClient=500000
ZENBytesRxClient=1200
ZENTotalBytesTxClient=7500000
ZENBytesTxClient=1500
ZENTotalBytesRxConnector=7500000
ZENBytesRxConnector=1500
ZENTotalBytesTxConnector=500000
ZENBytesTxConnector=1200
IdP=XYZ Azure AD
c2c=0
ClientCity=New York
MicroTenantID=0
AppMicroTenantID=0

This structured format ensures that logs are searchable, filterable, and ready for analysis within the SIEM. Security teams can use it to track user access, monitor data transfers, and investigate policy violations effectively.

How This Log Format Benefits SIEM Analysis

By parsing logs in this structured format, organizations can gain deeper insights into user behavior and security events. Each log entry becomes easier to search and correlate, allowing security teams to track application access, monitor potential security threats, and detect anomalies in real-time.

This format also enables faster incident response, as security analysts can quickly identify unauthorized access attempts, excessive data transfers, or unusual login behavior. It also integrates seamlessly with SIEM platforms, making it easier to generate reports, build dashboards, and conduct forensic investigations.

Creating Rules and Alerts in SIEM

Once logs are ingested into the SIEM, they can be used to define security rules and generate alerts for suspicious activities.